

Medvale Internal Medicine

Chapter 1G: Arrhythmias and Conduction Disease

Original study-text chapter for graph-RAG use. Copyrighted source images are not reproduced; original tables and text diagrams are used instead.

Why this matters clinically

Arrhythmias are disorders of impulse formation, impulse conduction, or both. They matter because the same symptom - palpitations, dizziness, dyspnea, chest discomfort, syncope, or fatigue - can represent a benign premature beat, atrial fibrillation with embolic stroke risk, pre-excited tachycardia that can degenerate into ventricular fibrillation, or ventricular tachycardia that signals structural heart disease. A safe clinician does not begin by memorizing rhythm names. The first step is to decide whether the patient is unstable, whether the QRS complex is narrow or wide, whether the rhythm is regular or irregular, and whether atrial activity can be identified.

This chapter is intentionally paragraph-heavy for graph-RAG use. Repeated associations are built into the prose: atrial fibrillation is linked to thromboembolism, left atrial appendage thrombus, rate control, rhythm control, anticoagulation, and cardioversion timing; ventricular tachycardia is linked to wide QRS, structural heart disease, myocardial scar, syncope, sudden cardiac death, amiodarone, cardioversion, defibrillation, and ICD consideration. The uploaded supplemental cardiology section emphasizes premature atrial and ventricular complexes, atrial fibrillation, atrial flutter, multifocal atrial tachycardia, PSVT, WPW, ventricular tachycardia, cardioversion/defibrillation distinctions, CHA2DS2-VASc stroke risk, and heart-failure device therapy associations.

Core first-pass rhythm framework

Every arrhythmia question can be simplified into a bedside sequence. First, determine whether the patient is hemodynamically unstable. Instability means hypotension, shock, ischemic chest pain, acute pulmonary edema, altered mental status, or syncope attributable to the rhythm. An unstable tachyarrhythmia generally requires synchronized cardioversion if the patient has a pulse. Pulseless ventricular tachycardia or ventricular fibrillation requires immediate unsynchronized defibrillation and CPR. Second, determine whether the QRS is narrow or wide. A narrow QRS rhythm usually originates above the ventricles and conducts through the His-Purkinje system. A wide QRS rhythm is ventricular until proven otherwise, especially in an older patient or a patient with prior MI, cardiomyopathy, heart failure, or structural heart disease. Third, determine whether the rhythm is regular or irregular. Regular narrow-complex tachycardia points toward AVNRT, orthodromic AVRT, or atrial flutter with fixed conduction. Irregularly irregular narrow-complex tachycardia strongly suggests atrial fibrillation, although multifocal atrial tachycardia can mimic it when P waves are present with varying morphology.

Medvale rule

In an adult with a wide-complex tachycardia, especially with prior MI or cardiomyopathy, treat the rhythm as ventricular tachycardia until proven otherwise. The uploaded source highlights the same practical rule: always suspect VT in a wide, greater-than-0.12-second QRS tachycardia.

Question	Bedside interpretation	Management implication
Is the patient unstable?	Hypotension, shock, ischemic chest pain, pulmonary edema, altered mental status, or syncope.	Unstable tachycardia with pulse -> synchronized cardioversion. VF/pulseless VT -> defibrillation.
Is QRS narrow or wide?	Narrow is usually supraventricular. Wide is VT, SVT with aberrancy, or pre-excited tachycardia.	Wide-complex tachycardia in structural heart disease should be presumed VT.
Is rhythm regular or irregular?	Regular suggests organized circuit. Irregularly irregular suggests AF or MAT.	Regular SVT may respond to vagal maneuvers or adenosine; AF requires rate/rhythm and stroke-risk decisions.
Are P waves visible?	Absent P waves may be AF, AVNRT, or buried atrial activity. Saw-tooth flutter waves suggest atrial flutter.	P-wave morphology can distinguish AF from MAT and flutter.

Numbers to know: arrhythmia anchors

Topic	Number or cutoff	Clinical association
Wide QRS	≥ 0.12 seconds / ≥ 120 ms	Wide-complex tachycardia should raise concern for VT.
Narrow QRS	< 120 ms	Usually supraventricular conduction through normal His-Purkinje system.
First-degree AV block	PR > 200 ms	Delayed AV conduction; often benign unless symptomatic or drug-related.
WPW pattern	Short PR often < 100 ms plus delta wave	Accessory pathway pre-excites ventricle.
AF atrial activity	> 400 atrial impulses/min in source	Chaotic atrial activity with irregularly irregular ventricular rhythm.
AF ventricular response	Often 75-175 bpm in source	Rate varies with AV nodal conduction and medications.
Lenient AF rate target	< 110 bpm	Reasonable in many stable patients when symptoms and LV function permit.
AF duration before cardioversion concern	> 48 hours or unknown	Anticoagulation/TEE strategy needed to reduce embolic risk.
Pre-cardioversion anticoagulation	3 weeks	Classic strategy if AF > 48 hr or unknown duration.
Post-cardioversion anticoagulation	At least 4 weeks	Atrial stunning and embolic risk persist after conversion.
Atrial flutter atrial rate	250-350 bpm	Saw-tooth flutter waves; often 2:1 or 3:1 conduction.
MAT diagnosis	At least 3 P-wave morphologies	Irregular rhythm often in severe pulmonary disease.
Wandering atrial pacemaker	Same P-wave variability but HR 60-100 bpm	MAT-like morphology without tachycardia.
VT rate in source	120-250 bpm	Three or more PVCs in a row from ventricular focus.
Sustained VT	> 30 seconds or requiring intervention	More dangerous; associated with hemodynamic compromise.
Stable VT systolic BP anchor	> 90 mm Hg in source	Stable mild symptoms may permit pharmacologic therapy.
HFrEF device threshold	EF $\leq 35\%$ in source	ICD/CRT considerations depend on symptoms, timing, QRS, and GDMT.

Premature atrial complexes

A premature atrial complex is an early atrial beat that arises outside the sinus node. Because the ectopic focus is still in the atrium, the QRS is usually narrow unless there is aberrant conduction. The P wave often looks different from the normal sinus P wave because atrial activation begins from an abnormal location. PACs can occur in healthy adults, especially with adrenergic stimulation, alcohol, caffeine, tobacco, sleep deprivation, hypoxia, infection, electrolyte abnormalities, or underlying atrial stretch. They are often asymptomatic, but patients may report a skipped beat, flutter, or brief palpitation. The uploaded source notes that PACs can be seen in more than 50% of normal adults during longer monitoring and that treatment is usually unnecessary unless symptoms are frequent or PACs trigger more sustained tachyarrhythmias.

The graph-RAG association for PACs is: early atrial focus $\rightarrow$ abnormal P-wave morphology $\rightarrow$ usually narrow QRS $\rightarrow$ often benign $\rightarrow$ may trigger PSVT or atrial fibrillation. PACs are not the same as atrial fibrillation. A PAC is an isolated premature atrial impulse; atrial fibrillation is continuous chaotic atrial activation with loss of organized atrial contraction.

Premature ventricular complexes

A premature ventricular complex is an early beat arising from a ventricular focus. Because the impulse begins below the His-Purkinje system and spreads through ventricular myocardium rather than the normal conduction pathway, the QRS is wide, bizarre, and usually followed by a compensatory pause. PVCs are common and may be benign in structurally normal hearts, but they become more concerning when they are frequent, symptomatic, multifocal, occur in runs, appear during or after myocardial infarction, or occur in patients with cardiomyopathy or reduced left ventricular function.

PVC patterns have specific names that are useful for exams and for graph retrieval. Couplet means two consecutive PVCs. Bigeminy means every other beat is a PVC. Trigeminy means every third beat is a PVC. Three or more ventricular beats in a row can be considered ventricular tachycardia if the rate is fast enough. The source emphasizes that frequent PVCs after MI can increase risk for sudden death, but suppressing PVCs with certain antiarrhythmic drugs after MI historically increased mortality in CAST-style trials. The practical lesson is not to treat every PVC reflexively; instead, evaluate the underlying heart, electrolytes, ischemia, medication triggers, and symptoms.

PVC concept	Meaning	Association
Isolated PVC	Single premature ventricular beat.	Often benign if no structural disease.
Couplet	Two PVCs in a row.	More ectopy burden than isolated PVC.
Bigeminy	Sinus beat followed by PVC repeatedly.	May cause palpitations or pseudo-bradycardia.
Trigeminy	Two sinus beats followed by PVC repeatedly.	Pattern recognition term.
Nonsustained VT	Run of VT that self-terminates before 30 sec.	Risk depends heavily on structural heart disease.
Sustained VT	Persists >30 sec or requires intervention.	Potentially life-threatening; consider ICD evaluation.

Atrial fibrillation

Atrial fibrillation is a chaotic atrial rhythm in which multiple atrial wavelets or drivers activate the atria continuously. The atria do not contract effectively, and the AV node receives many impulses irregularly. Because the AV node blocks some impulses and conducts others unpredictably, the ventricular rhythm becomes irregularly irregular. The ECG lacks organized P waves; instead, it may show a wavy or fibrillatory baseline with irregular R-R intervals. The uploaded source describes atrial rates above 400 per minute and ventricular rates that commonly fall between about 75 and 175 per minute, depending on AV nodal conduction, sympathetic tone, medications, and underlying disease.

The two most important clinical consequences of AF are hemodynamic compromise and thromboembolism. Hemodynamic compromise occurs because the patient loses coordinated atrial contraction, may develop a rapid ventricular response, and may have less diastolic filling time. This is especially important in heart failure, hypertrophic cardiomyopathy, severe aortic stenosis, mitral stenosis, or older patients with stiff ventricles who rely on atrial kick. Thromboembolism occurs because stagnant blood can form thrombus in the left atrial appendage. If a clot embolizes to the brain, the patient develops ischemic stroke. This is why AF management has three linked goals: control the ventricular rate, decide whether to restore sinus rhythm, and prevent embolic stroke when risk justifies anticoagulation.

Atrial fibrillation: acute management logic

The first question in AF is stability. A patient with AF and hypotension, shock, acute pulmonary edema, ischemic chest pain, or altered mental status attributable to the arrhythmia should undergo immediate synchronized cardioversion. If the patient is stable, rate control is usually the first step. A lenient rate target below about 110 beats/min is acceptable for many stable patients, although stricter control may be needed when symptoms persist, tachycardia-mediated cardiomyopathy is suspected, or heart failure worsens. Beta-blockers and non-dihydropyridine calcium channel blockers such as diltiazem or verapamil slow AV nodal conduction. In HFrEF, non-dihydropyridine calcium channel blockers can worsen systolic function, so beta-blockers, digoxin, or amiodarone may be considered depending on acuity and blood pressure.

Rhythm control means attempting to restore and maintain sinus rhythm using electrical cardioversion, antiarrhythmic medications, or ablation. It is reasonable when AF causes major symptoms, when this is the first episode, when rate control fails, when tachycardia-mediated cardiomyopathy is suspected, or in selected patients with heart failure. The uploaded source emphasizes that rate control has not been inferior to rhythm control in major older trial framing for many patients, but rhythm control remains appropriate for symptoms and selected clinical scenarios. Modern AF guidance also emphasizes shared decision-making, stroke-risk assessment, anticoagulation decisions, risk-factor modification, and ablation in selected symptomatic patients.

Atrial fibrillation: anticoagulation and cardioversion timing

Cardioversion can dislodge atrial thrombus, and atrial mechanical function can remain stunned even after sinus rhythm returns. Therefore, AF duration matters. If AF has lasted more than 48 hours or the duration is unknown, classic management is anticoagulation for 3 weeks before elective cardioversion and at least 4 weeks after cardioversion. An alternative is transesophageal echocardiography to look for left atrial thrombus; if no thrombus is present, cardioversion may proceed with appropriate anticoagulation, followed by at least 4 weeks of anticoagulation afterward. If AF duration is clearly less than 48 hours, the embolic risk is lower, but patient-specific stroke risk still matters.

AF problem	Core association	Management association
Rapid irregular rhythm	Irregularly irregular R-R intervals and absent organized P waves.	Rate control with AV nodal blocker if stable.
Hemodynamic instability	Hypotension, shock, pulmonary edema, ischemia, altered mental status.	Immediate synchronized cardioversion.
Stroke prevention	Left atrial appendage thrombus can embolize.	Use CHA2DS2-VASc-style risk assessment and anticoagulation when indicated.
AF >48 hr or unknown duration	Higher cardioversion embolic risk.	Anticoagulate 3 weeks before and at least 4 weeks after, or use TEE-guided strategy.
AF with HFrEF	Loss of atrial kick and tachycardia may worsen HF.	Avoid non-DHP CCBs; consider beta-blocker, digoxin, amiodarone, or rhythm strategy based on context.

CHA2DS2-VASc-style stroke-risk associations

Stroke risk in AF rises with age, heart failure, hypertension, diabetes, prior stroke or TIA, vascular disease, and female sex as a risk modifier. The uploaded source provides a table showing annual stroke risk rising as CHA2DS2-VASc score increases, from near general-population risk at 0 to markedly higher risk at scores such as 6, 8, or 9. Contemporary AF guidelines recommend using validated clinical risk scores, considering bleeding risk as a modifiable-safety tool rather than as a reason to ignore stroke risk, and preferring direct oral anticoagulants over warfarin in many patients without mechanical valves or moderate-to-severe rheumatic mitral stenosis.

CHA2DS2-VASc component	Points	Why it matters
Congestive heart failure / LV dysfunction	1	Atrial stretch and poor flow increase thromboembolic risk.
Hypertension	1	Common stroke-risk driver.
Age 65-74	1	Risk rises with age.
Age ≥75	2	Older age carries stronger risk weight.
Diabetes mellitus	1	Vascular and thrombotic risk.
Stroke/TIA/systemic embolism history	2	Strongest clinical risk marker.
Vascular disease	1	CAD, PAD, or aortic plaque reflect systemic vascular risk.
Sex category	1 for female sex in scoring systems	Interpreted with other risk factors in modern decision-making.

Atrial flutter

Atrial flutter is an organized macroreentrant atrial tachycardia, often with atrial rates around 250-350 beats/min. The atria activate rapidly but regularly, and the AV node often conducts every second or third flutter wave, creating 2:1 or 3:1 conduction. The classic ECG clue is a saw-tooth baseline, best seen in inferior leads II, III, and aVF. Because atrial flutter and AF share thromboembolic mechanisms, anticoagulation decisions are generally made using the same stroke-risk logic as AF. Treatment also overlaps: rate control, rhythm control, anticoagulation when indicated, and catheter ablation for recurrent or symptomatic flutter. The source emphasizes that flutter ablation is highly successful and may avoid long-term rhythm-drug exposure, although anticoagulation still depends on risk factors and recurrence potential.

Multifocal atrial tachycardia

Multifocal atrial tachycardia is an irregular atrial tachycardia caused by multiple atrial pacemaker sites. It is strongly associated with severe pulmonary disease, especially COPD, hypoxemia, and acute respiratory illness. The ECG shows an irregular rhythm with at least three different P-wave morphologies, variable PR intervals, and variable R-

R intervals. This can be confused with AF because both rhythms are irregular, but MAT has visible P waves with changing shapes, whereas AF lacks organized P waves. The source notes that wandering atrial pacemaker has the same type of variable P-wave morphology but occurs at a heart rate between 60 and 100 beats/min rather than a tachycardic rate.

Management of MAT focuses on the underlying cause: improve oxygenation and ventilation, treat COPD exacerbation or pneumonia, correct potassium and magnesium abnormalities, and reduce adrenergic triggers. Electrical cardioversion is ineffective because MAT is not a single reentrant circuit that can be reset by a shock. This is a high-yield distinction: atrial flutter and many PSVTs are circuit-driven and can be terminated; MAT is multifocal automaticity and requires physiologic correction.

Paroxysmal supraventricular tachycardia: AVNRT and AVRT

Paroxysmal supraventricular tachycardia is a sudden-onset, sudden-offset tachycardia that usually uses a reentrant circuit. The most common mechanism is AV nodal reentrant tachycardia, in which two pathways within or near the AV node - one fast and one slow - allow a loop of electrical activity to perpetuate itself. AVNRT typically produces a regular narrow-complex tachycardia, often with P waves hidden inside the QRS or appearing immediately after it. Orthodromic AV reentrant tachycardia uses an accessory pathway between atrium and ventricle, with the impulse going down the AV node and returning retrograde through the accessory pathway. Because ventricular activation still uses the normal His-Purkinje system in orthodromic AVRT, the QRS is usually narrow.

Stable regular narrow-complex SVT is treated first with vagal maneuvers such as the Valsalva maneuver. If vagal maneuvers fail, adenosine is the classic acute medication because it transiently blocks AV nodal conduction and can terminate AV-node-dependent reentry. Adenosine can also help reveal underlying atrial flutter waves by slowing AV nodal conduction. If the patient is unstable, synchronized cardioversion is indicated rather than prolonged medication trials. The 2015 ACC/AHA/HRS SVT guideline recommends vagal maneuvers and adenosine as acute treatment options for regular SVT, with synchronized cardioversion for unstable or refractory cases.

Original rhythm decision diagram

Tachycardia with a pulse

```
|
|-- Unstable? -> synchronized cardioversion
|
|-- Stable -> check QRS width
|       |
|       |-- Narrow and regular -> vagal maneuvers -> adenosine
|       |
|       |-- Narrow and irregular -> AF, flutter with variable block, or MAT
|       |
|       |-- Wide and regular -> presume VT, especially with structural heart disease
|       |
|       |-- Wide and irregular -> AF with aberrancy, pre-excited AF, polymorphic VT, or torsades
```

Wolff-Parkinson-White syndrome and pre-excitation

Wolff-Parkinson-White syndrome involves an accessory pathway that bypasses the AV node and connects atrium to ventricle. On a baseline ECG, the ventricle is activated early, producing a short PR interval, often less than 100 ms in the uploaded source, and a slurred upstroke of the QRS called a delta wave. WPW can produce orthodromic AVRT, which is usually narrow-complex, or antidromic AVRT, which is wide-complex because ventricular activation proceeds through the accessory pathway. The most dangerous scenario is atrial fibrillation with WPW, because chaotic atrial impulses can bypass the AV node and conduct rapidly to the ventricles, potentially degenerating into ventricular fibrillation.

The treatment distinction is essential. AV nodal blockers such as adenosine, beta-blockers, nondihydropyridine calcium channel blockers, and digoxin can be dangerous in pre-excited AF because blocking the AV node may promote conduction down the accessory pathway. For stable pre-excited AF, drugs that slow accessory pathway

conduction, such as procainamide in many teaching algorithms, are preferred; unstable patients require synchronized cardioversion. Definitive therapy for symptomatic WPW is catheter ablation of the accessory pathway.

WPW trap

Do not treat irregular wide-complex tachycardia with suspected pre-excitation like ordinary AF. AV nodal blockade can worsen accessory pathway conduction and increase ventricular fibrillation risk.

Ventricular tachycardia

Ventricular tachycardia is a rapid rhythm arising below the bundle of His. It is often defined as three or more consecutive ventricular beats at a tachycardic rate, and sustained VT lasts more than 30 seconds or requires termination because of symptoms or instability. VT is strongly associated with structural heart disease, especially prior myocardial infarction and scar. The scar creates regions of slow conduction that allow reentry. VT can also occur in active ischemia, cardiomyopathy, myocarditis, infiltrative disease, electrolyte disturbances, drug toxicity, congenital channelopathies, and prolonged QT syndromes.

Clinically, VT may cause palpitations, dyspnea, chest pain, lightheadedness, syncope, hypotension, or sudden cardiac death. The ECG usually shows a wide QRS tachycardia. AV dissociation, capture beats, fusion beats, extreme axis, and concordance support VT, but in emergency practice the safest rule is simpler: a regular wide-complex tachycardia in an adult with structural heart disease is VT until proven otherwise. The source repeatedly emphasizes wide and bizarre QRS complexes and the practical cutoff that QRS greater than 0.12 seconds should raise VT concern.

Treatment depends on pulse and stability. Pulseless VT is treated like ventricular fibrillation with CPR and defibrillation. Unstable VT with a pulse is treated with synchronized cardioversion. Stable monomorphic VT may be treated pharmacologically, commonly with IV amiodarone in many teaching algorithms, while reversible causes such as ischemia, hypokalemia, hypomagnesemia, drug toxicity, and QT prolongation are corrected. Sustained VT usually warrants evaluation for structural heart disease and consideration of ICD therapy unless the cause is clearly reversible and unlikely to recur.

Torsades de pointes and polymorphic VT

Torsades de pointes is a polymorphic VT associated with prolonged QT interval. Risk factors include hypokalemia, hypomagnesemia, hypocalcemia, bradycardia, congenital long-QT syndromes, and many medications such as class IA and class III antiarrhythmics, certain antibiotics, antipsychotics, and antidepressants. The ECG appears to twist around the baseline. Management centers on removing QT-prolonging drugs, correcting potassium and magnesium, giving IV magnesium, and increasing heart rate in recurrent pause-dependent torsades when appropriate. Torsades is dangerous because it can degenerate into ventricular fibrillation.

VT concept	Key clue	Management association
Sustained monomorphic VT	Regular wide-complex tachycardia lasting >30 sec or requiring intervention.	Cardioversion if unstable; antiarrhythmic if stable; evaluate for ICD.
Pulseless VT	No pulse with organized wide-complex VT pattern.	Defibrillation + CPR.
Polymorphic VT	Changing QRS morphology.	Treat ischemia, electrolytes, QT/drug causes; defibrillate if unstable/pulseless.
Torsades de pointes	Polymorphic VT with prolonged QT.	IV magnesium, stop QT-prolongers, correct K/Mg, consider pacing/isoproterenol in select cases.
Nonsustained VT	Self-terminating run <30 sec.	Risk depends on CAD, EF, symptoms, and structural disease.

Cardioversion versus defibrillation

Cardioversion and defibrillation both deliver electrical energy, but they are used differently. Synchronized cardioversion times the shock with the QRS complex to avoid delivering energy during the vulnerable repolarization period. It is used for unstable tachyarrhythmias with a pulse, such as unstable AF, atrial flutter, SVT, or monomorphic VT. Defibrillation is unsynchronized and is used for rhythms without an organized perfusing pulse,

especially ventricular fibrillation and pulseless VT. The uploaded source includes a clinical pearl that emphasizes this distinction: cardioversion is synchronized with the QRS; defibrillation is not synchronized and is used to convert life-threatening rhythms to normal rhythm.

Bradyarrhythmias and AV block

Although the uploaded supplemental chapter emphasizes tachyarrhythmias, internal medicine cardiology also requires a bradycardia framework. Sinus bradycardia can be normal in athletes or during sleep, but it may also result from inferior MI, medications such as beta-blockers, calcium channel blockers, digoxin, or amiodarone, hypothyroidism, hyperkalemia, increased vagal tone, or intrinsic sinus node disease. The management question is whether the patient is symptomatic and whether the bradycardia is causing hypotension, syncope, ischemia, or heart failure.

AV block is classified by the relationship between P waves and QRS complexes. First-degree AV block has a PR interval greater than 200 ms with every P wave conducted. Mobitz I second-degree block has progressive PR prolongation before a dropped beat and is often AV nodal. Mobitz II has sudden dropped QRS complexes without progressive PR prolongation and is usually infranodal, making it more dangerous and more likely to require pacing. Third-degree AV block has complete AV dissociation: atria and ventricles beat independently. In MI, inferior infarcts may produce AV nodal block that can be transient, while anterior infarcts can damage the His-Purkinje system and carry a worse prognosis.

Conduction disorder	ECG clue	High-yield implication
First-degree AV block	PR >200 ms, all P waves conducted.	Usually observe unless symptomatic or medication-related.
Mobitz I	Progressive PR lengthening then dropped beat.	Often AV nodal; may be vagal or inferior MI related.
Mobitz II	Dropped QRS without progressive PR prolongation.	Infranodal; pacing often needed.
Third-degree AV block	AV dissociation.	Unstable patients need urgent pacing; evaluate reversible causes.
Sinus node dysfunction	Sinus pauses, bradycardia-tachycardia pattern.	Pacemaker if symptomatic and not reversible.

Device therapy associations: pacemakers, ICDs, and CRT

Pacemakers treat symptomatic bradycardia or conduction disease. They do not prevent ventricular fibrillation unless paired with defibrillator capability. Implantable cardioverter-defibrillators detect malignant ventricular arrhythmias and terminate them with antitachycardia pacing or shock. ICDs are used for secondary prevention after survived cardiac arrest or sustained VT not due to a reversible cause, and for primary prevention in selected patients with reduced EF after guideline-directed therapy and appropriate waiting periods. The uploaded source connects ICD benefit to prevention of sudden cardiac death in high-risk heart failure patients and describes EF $\leq 35\%$ with symptomatic heart failure as a classic anchor.

Cardiac resynchronization therapy is a biventricular pacing strategy for selected patients with HFrEF and electrical dyssynchrony, often wide QRS with left bundle branch block morphology. CRT improves the timing of ventricular contraction, which can improve symptoms and reduce heart-failure events in appropriate patients. Ventricular assist devices are mechanical pumps for advanced heart failure, either as bridge to transplant, bridge to recovery, bridge to decision, or destination therapy. Although VADs are not arrhythmia treatments in the narrow sense, advanced HF, ventricular arrhythmias, ICD decisions, CRT decisions, and VAD decisions are clinically connected.

Antiarrhythmic medication framework

Antiarrhythmic drugs are useful but dangerous because many can provoke arrhythmias, worsen heart failure, slow conduction excessively, or interact with other medications. A practical internal medicine framework is more useful than memorizing every drug. Class I drugs block sodium channels and can slow conduction; they are generally avoided in significant structural heart disease unless specifically indicated by electrophysiology expertise. Class II drugs are beta-blockers and are often useful in ischemic heart disease, rate control, and adrenergic arrhythmias. Class III drugs prolong repolarization; amiodarone is effective for many atrial and ventricular arrhythmias but has

thyroid, pulmonary, hepatic, ocular, skin, and neurologic toxicities. Class IV drugs are nondihydropyridine calcium channel blockers, such as diltiazem and verapamil, which slow AV nodal conduction but can worsen HFrEF.

Drug/class	Useful association	Caution
Adenosine	Terminates AV-node-dependent SVT; reveals flutter waves.	Avoid/caution in severe bronchospasm; do not use for irregular wide pre-excited AF.
Beta-blockers	AF rate control, ischemic arrhythmias, adrenergic states.	Bradycardia, AV block, shock, decompensated HF.
Diltiazem/verapamil	AF/flutter rate control if EF preserved.	Avoid in HFrEF or shock.
Digoxin	AV nodal slowing, sometimes useful in HFrEF with AF.	Narrow therapeutic index; toxicity causes GI, visual, CNS, and arrhythmia findings.
Amiodarone	AF rhythm control in selected patients; VT treatment.	Pulmonary, thyroid, liver, skin, ocular, neurologic toxicity; long half-life.
Procainamide	Selected stable wide-complex tachycardias and pre-excited AF.	Hypotension, QRS widening, lupus-like syndrome with chronic use.
Magnesium	Torsades de pointes.	Monitor renal function and reflexes in high doses.

High-yield rhythm illness scripts

Rhythm	ECG pattern	Classic association
PAC	Early abnormal P wave, usually narrow QRS.	Benign ectopy; may trigger SVT/AF.
PVC	Early wide bizarre QRS with compensatory pause.	Structural disease risk depends on burden/context.
AF	Irregularly irregular, no organized P waves.	Stroke risk, rate/rhythm control, anticoagulation.
Atrial flutter	Saw-tooth flutter waves, often 2:1 conduction.	Similar anticoagulation logic to AF; ablation effective.
MAT	Irregular rhythm with ≥ 3 P-wave morphologies.	COPD/hypoxemia; cardioversion ineffective.
AVNRT	Regular narrow SVT, P waves hidden or retrograde.	Vagal maneuvers, adenosine, AV nodal blockers, ablation.
Orthodromic AVRT	Regular narrow SVT using accessory pathway retrograde.	Adenosine can terminate if stable; ablation definitive.
WPW pattern	Short PR, delta wave, wide-ish QRS.	Avoid AV nodal blockers in pre-excited AF.
VT	Wide-complex tachycardia, often regular.	Prior MI/scar, cardiomyopathy, sudden death risk.
Torsades	Polymorphic VT with long QT.	Magnesium, correct electrolytes, stop QT drugs.

Common pitfalls

One common pitfall is treating a wide-complex tachycardia as SVT with aberrancy before excluding VT. In adults with prior MI or cardiomyopathy, VT is much more dangerous and must be presumed until proven otherwise. Another pitfall is giving AV nodal blockers to a patient with irregular wide-complex tachycardia from pre-excited AF. In that setting, blocking the AV node can shift conduction toward the accessory pathway and precipitate ventricular fibrillation. A third pitfall is forgetting anticoagulation around cardioversion when AF has lasted more than 48 hours or duration is unknown. A fourth pitfall is trying to electrically cardiovert MAT, which is usually driven by pulmonary disease and multifocal atrial automaticity rather than one reentrant circuit. A fifth pitfall is suppressing benign PVCs with antiarrhythmic drugs without evaluating the underlying heart; the risk and treatment depend on structural disease, symptoms, ischemia, electrolyte abnormalities, and PVC burden.

Practice questions

Question 1

Stem: A 74-year-old man with prior MI develops a regular wide-complex tachycardia at 180 beats/min. Blood pressure is 78/44 and he is confused. What is the next step?

Answer and teaching point: Treat as unstable VT with a pulse and perform synchronized cardioversion. A wide-complex tachycardia in structural heart disease is VT until proven otherwise.

Question 2

Stem: A stable patient has palpitations. ECG shows an irregularly irregular rhythm without organized P waves. Duration is unknown. What two management decisions are central?

Answer and teaching point: Control rate or choose rhythm strategy, and assess need for anticoagulation. If elective cardioversion is planned and duration is >48 hours or unknown, use anticoagulation for 3 weeks before and at least 4 weeks after, or use a TEE-guided strategy.

Question 3

Stem: A patient with severe COPD has tachycardia with an irregular rhythm. ECG shows at least three distinct P-wave morphologies and variable PR intervals. What is the rhythm and treatment focus?

Answer and teaching point: Multifocal atrial tachycardia. Treat the underlying pulmonary disease, improve oxygenation/ventilation, and correct electrolytes. Cardioversion is ineffective.

Question 4

Stem: A young patient has recurrent sudden palpitations. ECG during symptoms shows regular narrow-complex tachycardia with no visible P waves. Vagal maneuvers fail. What medication is classically used acutely if stable?

Answer and teaching point: Adenosine. It transiently blocks AV nodal conduction and can terminate AVNRT or orthodromic AVRT.

Question 5

Stem: A baseline ECG shows a short PR interval and delta wave. Later the patient develops irregular wide-complex tachycardia. Which medication class should be avoided?

Answer and teaching point: Avoid AV nodal blockers such as adenosine, beta-blockers, diltiazem/verapamil, and digoxin when pre-excited AF is suspected.

Question 6

Stem: A patient has polymorphic VT with prolonged QT after starting a QT-prolonging drug. What treatment principles apply?

Answer and teaching point: Stop the QT-prolonging drug, correct potassium and magnesium, give IV magnesium, and defibrillate if unstable or pulseless.

Chapter summary

Arrhythmia management begins with stability, QRS width, rhythm regularity, and P-wave assessment. Premature atrial complexes and PVCs are common ectopic beats, but their significance depends on symptoms and structural heart disease. Atrial fibrillation is an irregularly irregular rhythm without organized P waves and requires decisions about rate control, rhythm control, and anticoagulation. Atrial flutter is organized macroreentry with saw-tooth flutter waves and atrial rates around 250-350 beats/min. MAT is an irregular atrial tachycardia with at least three P-wave morphologies and is strongly linked to COPD and hypoxemia. PSVT, especially AVNRT and orthodromic AVRT, is usually regular, narrow, sudden-onset, and AV-node-dependent; vagal maneuvers and adenosine are classic acute treatments in stable patients. WPW creates pre-excitation with short PR and delta wave, and AV nodal blockers can be dangerous in pre-excited AF. Ventricular tachycardia is a wide-complex rhythm arising below the His bundle, often linked to prior MI, scar, cardiomyopathy, and sudden cardiac death. Sustained VT lasts more than 30 seconds or requires intervention; unstable VT requires synchronized cardioversion, and pulseless VT/VF requires defibrillation. Device therapy connects rhythm disease to heart failure: pacemakers treat bradycardia, ICDs prevent sudden death from malignant ventricular arrhythmias, and CRT can help selected HFrEF patients with dyssynchrony.

Source and reference notes

Coverage checklist: user-uploaded cardiology4 supplemental chapter, especially pages 3-20, which cover PACs, PVCs, AF, flutter, MAT, PSVT, WPW, VT, cardioversion/defibrillation, anticoagulation concepts, and device therapy associations. Current guideline framing was cross-checked against the 2023 ACC/AHA/ACCP/HRS atrial fibrillation guideline, the 2015 ACC/AHA/HRS supraventricular tachycardia guideline, and the 2017 AHA/ACC/HRS ventricular arrhythmia/sudden death guideline. This textbook section is for medical education and does not replace clinical protocols, institutional pathways, or specialist consultation.